

Smart Farming Robot

Soil sensing rover, canopy vision and budgeted hysteresis irrigation

AT A GLANCE

77.0 %

Water saved vs timer

15 / 266

Stress-hours smart / timer

96.0 %

Vision accuracy (held-out)

0.0121

Probe RMSE (VWC)

2488 L

Water used (3 days)

0.61 kWh

Pump energy

18/18

Unit tests

VERDICT

SIM PASS

TESTS 18/18

VISION OK

DECLARED SUBSTITUTION

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1. Objective and brief mapping

The brief asks for a field model with crop plots, sensor simulation with calibration/noise/dropout, a rover that covers the field, image-based crop-health classification, an automated irrigation controller with hysteresis, a daily budget and priority scheduling, a comparison against a timer schedule, and an analytics report. I implemented every item as a pure-Python simulation (numpy, OpenCV, scikit-learn, matplotlib) in **src/farm/** so that the same modules are imported by the standalone simulator, the unit tests and the ROS 2 node.

The 12 x 8 field (96 plots of 2 m x 2 m, 0.30 m root zone, so 1200 L raises one plot by 1.0 VWC) was simulated for 3 days at a 15-minute step with a kharif-season diurnal temperature profile (22-36 degC), Hargreaves-style evapotranspiration, 1 random rain event (8.4 mm), a dry patch (blocked drip line) and a leaf-spot patch seeded in two corners so the controller and the vision pipeline have something to find.

2. Architecture

```
Field 8x12: moisture, temperature, growth, health, stress_hours
    <-- ET(T_air, solar, crop coeff), rain events, irrigation litres per plot
Rover (boustrophedon, 96 plots, 30 plots/h)
    --probe--> CapacitiveMoistureSensor: ADC curve, 18-count noise, 3 % dropout
    --1-Wire--> DS18B20: 12-bit, +/-0.5 degC bias, 2 % dropout
    --camera--> render_canopy(health, growth) -> RGB + NIR
                -> ExG / VARI / pseudo-NDVI, leaf area, HSV brown spots + morphology
                -> DecisionTree(depth 4) -> healthy / stressed / diseased -> alerts
MoistureObserver (probe reset + dead-reckoned water balance) --est-->
HysteresisController: on < 0.20, off > 0.27 VWC, 1200 L/day budget,
    priority = deficit + stress history, pump 25 L/min, max 4 valves
Analytics: CSV logs, dashboard.png, summary.json/txt, frames -> farm_sim.gif
```

3. Implementation details

- **Sensors.** The capacitive probe follows the exponential-saturation ADC curve of the 3.3 V probes shipped with ESP32 kits (3100 counts in air, 1250 saturated); the firmware-side inverse is exact to 1e-6. Gaussian noise of 18 counts and 3 % dropouts are added, plus a 2 % gain error on the rover's probe; a median-of-3 read rejects dropouts. The DS18B20 model quantises to 0.0625 degC with a per-device bias in +/-0.5 degC.
- **Rover.** Boustrophedon path (95 unit moves for 96 plots), 30 plots/h, so one sweep takes 3.2 h; over 3 days it completed 22.5 sweeps and 6912 probe reads with 207 dropouts. Measured probe RMSE against the hidden truth was 0.0121 VWC with a -0.0114 bias from the uncalibrated gain.
- **Observer.** Because a plot is re-measured only every 3.2 h, the controller dead-reckons the soil-water balance between visits (nominal ET loss, plus litres it applied, plus rain-gauge input) and resets to the probe reading when fresh; its mean error was 0.0126 VWC. Without this the latched valve over-watered stale plots by thousands of litres in my first run - a real lesson about acting on old sensor data.
- **Vision.** Canopy images (96 x 96 RGB + NIR) are rendered from the hidden health/growth: leaf ellipses whose colour slides from green to chlorotic yellow, necrotic brown circles whose density grows as health falls, bluish-green weed rosettes and a 0.75-1.25 lighting gain. Features: mean ExG, VARI and pseudo-NDVI over the ExG vegetation mask, leaf-area fraction, HSV brown-spot count/area after open/close morphology and connected components, mean hue and weed fraction. A depth-4 decision tree is trained on 240 rendered plots; hand-tuned rules (spot fraction > 2 % or NDVI < 0.20 -> diseased; NDVI < 0.50 -> stressed) serve as a fallback.
- **Irrigation.** Per-plot hysteresis (open below 0.20, close above 0.27 VWC) with the valve state latched in between so the drip line does not chatter; a 1200 L/day budget with priority = moisture deficit x 100 + 0.05 x stress-hours; at most 4 valves open on a 25 L/min, 370 W monoblock pump; the run for each plot is cut short at the litres needed to reach the off threshold. The baseline timer runs every plot for 3 min at 06:00 and 18:00 (18.75 L each), the way most small drip installations are set.

Sensing metric (3-day run)	Measured
Moisture probe reads / dropouts	6912 / 207 (3.0 %)
DS18B20 reads / dropouts	4608 / 83 (1.8 %)
Probe RMSE / bias vs hidden truth	0.0121 / -0.0114 VWC
Observer mean abs. error (between visits)	0.0126 VWC
Rover sweeps / distance	22.5 sweeps / 2448 plot-lengths (4.9 km)

4. Verification

All 18 unittest cases pass (sim/test_log.txt). The held-out vision confusion matrix on 150 freshly rendered plots (rows = truth, columns = predicted healthy / stressed / diseased) is [[45, 0, 0], [4, 47, 0], [0, 2, 52]], i.e. 96.0 % for the tree and 95.3 % for the rules; every disease case was flagged as at least stressed. Inside the 3-day run the tree classified 2304 rover images with 98.3 % accuracy and the final health map matched the hidden truth on 100 % of plots.

Group	What is checked	Tests	Result
Sensors	calibration monotonic + invertible; noise/dropout statistics; DS18B20 0.0625 degC quantisation; median-of-N rejects dropouts	4	PASS

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Group	What is checked	Tests	Result
Field / rover	ET dries and irrigation wets soil; stress accumulates below 0.18 VWC; boustrophedon covers 96 plots with unit steps; one sweep samples all plots	4	PASS
Vision	ExG/VARI/pseudo-NDVI/leaf-area on known images; spot detector counts 3 painted spots; health ordering; held-out accuracy ≥ 0.90	4	PASS
Irrigation	hysteresis open/close sequence; daily budget never exceeded and resets; priority serves lowest VWC first; observer water balance; timer fires 2x/day	5	PASS
End-to-end	3-day run: stress fraction $< 2\%$ (measured 0.22 %), less water and fewer stress-hours than timer, budget honoured	1	PASS

5. Results

The smart controller used 77.0 % less water and produced 94 % fewer stress-hours. The timer both wastes water (it pushed 500 L past field capacity in already-wet plots) and starves the dry patch, where one plot sat below the stress threshold for 38.5 h; the smart controller served that patch first and its worst plot saw 2.0 h of stress. The budget bound on 31 control steps late in the day, which the alert log records once per day. At roughly Rs 7/kWh the pump energy difference is small (about Rs 14 over 3 days); the real saving is the 8312 L of borewell water.

Metric (3 days, 96 plots)	Smart (hysteresis + budget)	Timer (2 x 3 min/day)
Water applied	2488 L	10800 L
Water drained below root zone	0 L	500 L
Crop stress-hours (sum over plots)	15.0 h	266.5 h
Worst single plot stress	2.0 h	38.5 h
Final minimum VWC	0.225	0.230
Final mean health	0.887	0.881
Pump run time / energy	100 min / 0.61 kWh	432 min / 2.66 kWh
Valve open events / budget-limited steps	26 / 31	n/a

Smart Farming Robot — 3-day analytics dashboard

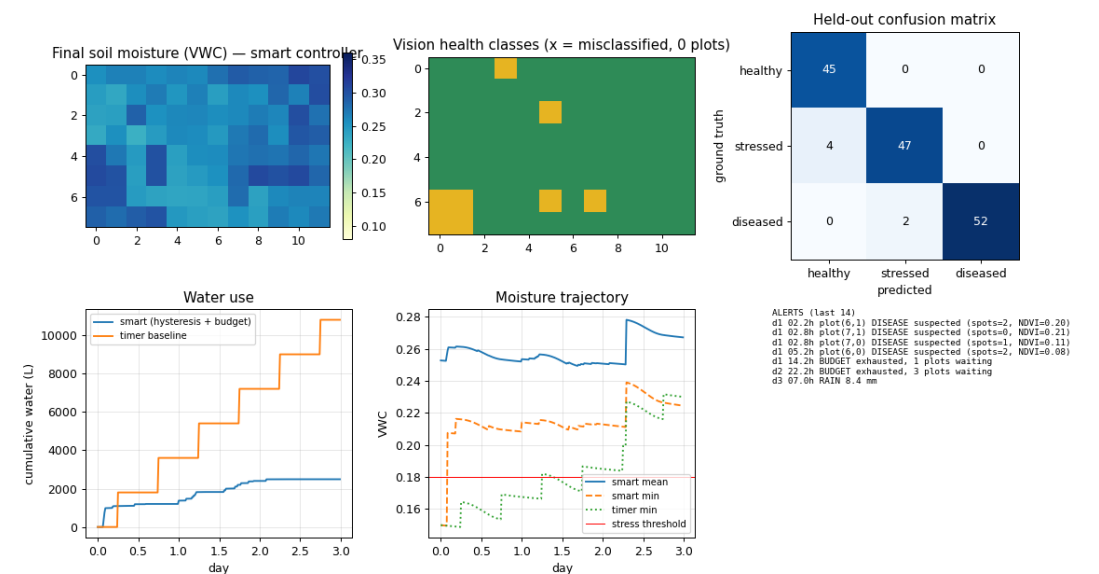


Figure 1 - dashboard.png from the run: final moisture, vision health map with misclassified plots, held-out confusion matrix, cumulative water, moisture trajectories, alert log.

5b. Per-day breakdown and example imagery

Day	Smart water	Timer water	Smart stress	Timer stress	Min VWC smart / timer	Rain
Day 1	1200 L	3600 L	15.0 h	192.0 h	0.150 / 0.149	0.0 mm
Day 2	1200 L	3600 L	0.0 h	74.5 h	0.210 / 0.166	0.0 mm
Day 3	88 L	3600 L	0.0 h	0.0 h	0.211 / 0.184	8.4 mm

Day 1 carries the cost of refilling the dry patch and the wetter half of the field draining to the on-threshold; from day 2 the smart controller settles to roughly the crop's evapotranspiration demand while the timer keeps applying its fixed 3600 L/day regardless of the rain on day 3. The daily minimum VWC shows where each strategy leaves its worst plot.

sim/alerts.txt (complete):

```
d1 02.2h plot(6,1) DISEASE suspected (spots=2, NDVI=0.20)
d1 02.8h plot(7,1) DISEASE suspected (spots=0, NDVI=0.21)
d1 02.8h plot(7,0) DISEASE suspected (spots=1, NDVI=0.11)
d1 05.2h plot(6,0) DISEASE suspected (spots=2, NDVI=0.08)
d1 14.2h BUDGET exhausted, 1 plots waiting
d2 22.2h BUDGET exhausted, 3 plots waiting
d3 07.0h RAIN 8.4 mm
```

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06

Robotics & Automation
Codec Technologies India
Internship 2026

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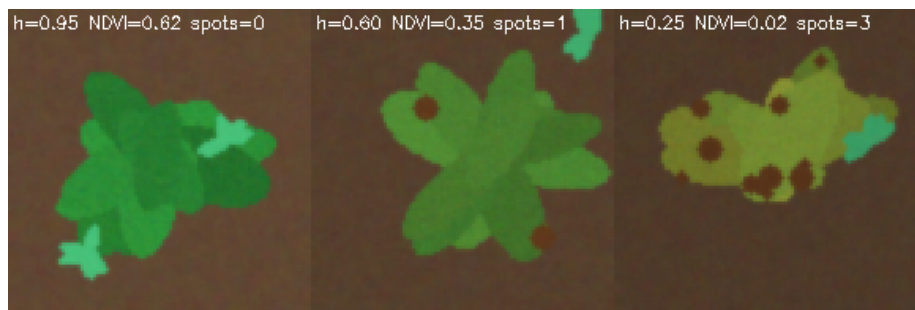
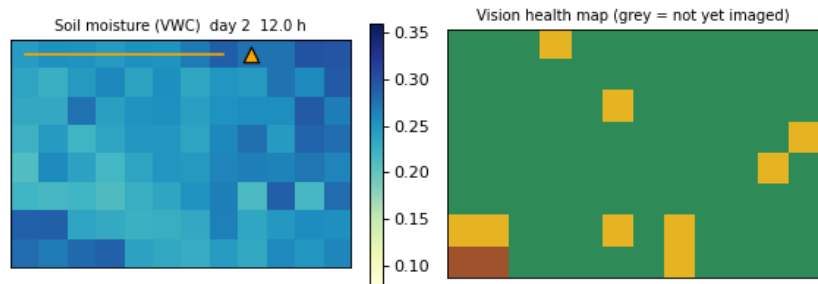


Figure 2 - canopy_examples.png: rendered plots at health 0.95 / 0.60 / 0.25 with the NDVI and brown-spot count the vision pipeline measured on them.

5c. Rover sweep frame



water used 1824 L stressed plots 0 rover battery 99%

Figure 3 - mid-run frame from farm_sim.gif: moisture heatmap with the rover (triangle), its recent boustrophedon trail, open valves (red squares) and the vision health map filling in as plots are imaged.

6. Declared substitutions / not run here

- **Webots:** worlds/farm_field.wbt (26 x 18 m arena, 4-wheel rover with canopy Camera, downward DistanceSensor used as the moisture probe, GPS + IMU) and worlds/farm_rover_controller.py are provided but were NOT run - Webots is not installed in the build container. Open the world in Webots R2023b with ../src on PYTHONPATH.
- **ROS 2 Humble:** ros2_ws/src/smart_farming_robot (package.xml, setup.py, node.py publishing /farm/moisture_map, /farm/health_map, /farm/valve_cmd, /farm/alerts, /rover/cmd_vel and subscribing to /soil/moisture_raw, /soil/temperature, /camera/image_raw; launch file) compiles with py_compile but was NOT launched - no rclpy here. Build with colcon build --packages-select smart_farming_robot, then ros2 launch smart_farming_robot smart_farming_robot.launch.py.
- Everything else - field dynamics, sensors, rover, vision, controller, analytics, GIF, tests - was run in this container with Python 3.11; every number above is read from sim/summary.json and sim/test_log.txt by docs/build_report.py.

7. How to run

```
cd 06_Smart_Farming_Robot
./run_all.sh                # sim + 18 unit tests, PASS/FAIL summary, exit code
make sim                    # -> sim/summary.json, dashboard.png, farm_sim.gif, CSV
                             logs
make test                   # python3 -m unittest discover -s tests -v
make report                 # docs/build_report.py -> tools/make_report.py ->
                             06_Report.pdf
cd src && python3 -m farm.simulate --days 5 --seed 3 --out ../sim # other scenarios
```

8. Which image features the classifier actually used

The tree leans on the RGB-only indices (ExG, VARI) far more than on the pseudo-NDVI I expected to dominate - chlorosis shifts the visible chromaticity strongly enough that the NIR channel adds little, which is good news for a rover with an ordinary Rs 1500 RGB camera.

Feature	Gini importance	Meaning
exg	0.549	excess green
vari	0.427	VARI
ndvi	0.016	pseudo-NDVI over canopy mask
hue	0.009	mean canopy hue

9. Learnings / next steps

Two things surprised me. First, the stale-sensor problem: with a 3.2 h revisit time a latched valve is dangerous, and a simple water-balance observer fixed it. Second, the decision tree only beats the hand rules by a few points; the rules are more explainable for a farmer-facing app. Next steps: replace the deterministic boustrophedon with an information-driven route that revisits plots near the on-threshold sooner, add weather-forecast-aware budgeting (skip irrigation when rain is likely), and validate the ADC calibration curve against a real probe in a pot of known VWC.